Universität Basel

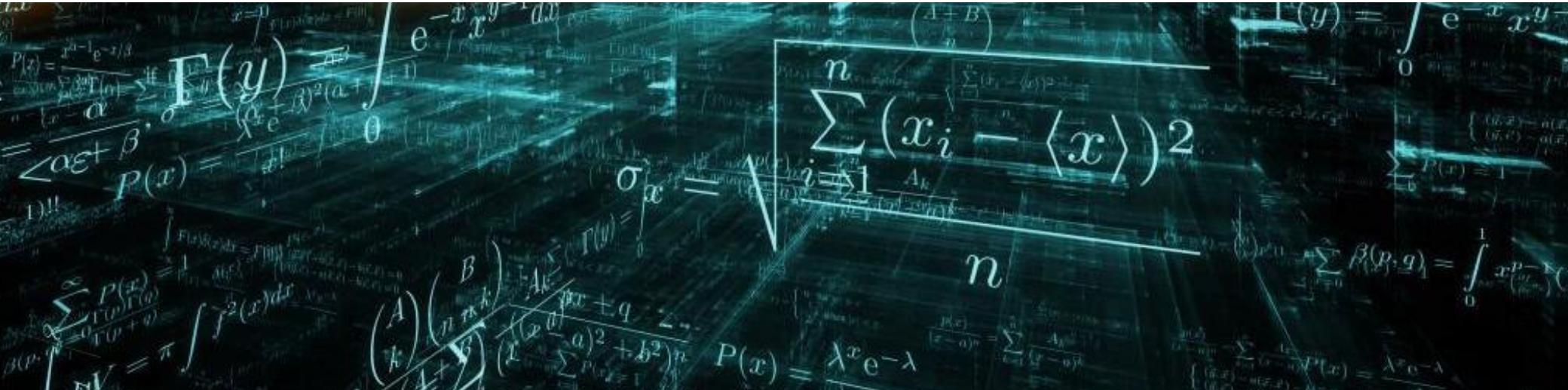
Computer Science / 15731-01 / 2026

Multimedia Retrieval

Chapter 0: Overview

Dr. Roger Weber, roger.weber@gmail.com

Admin
Motivation
How search works
Schedule



Admin

- **Time**: Friday 15:15 - 18:00
  (1st/2nd hour for theory, 3rd hour for exercises)
- **Location**: Spiegelgasse 5, Seminarraum 05.002
  Some instance will be online via Zoom
- **Exam**: Friday, Jan 8th, 2027, 14:00 - 16:00
  Biozentrum, Hörsaal U1.131
  - written exam, details will be discussed mid-semester and during the final lecture
  - **Note: Exam is immediately after the semester and new year break. Learning during the semester is essential to keep up**
- **Exercises**: every week, a new (optional) exercise is provided. It consists of 3 tasks
  - Quiz: multiple choice questions similar to what you can expect for the exam
  - Questions: free text answer similar to exam questions (but some may be bigger than at the exam)
  - Coding: deeper dive to implement retrieval aspects (it is explicitly encouraged to us AI tool to code)
  - Solutions are provided for self-correction, no hand-in of exercises, no corrections of answers

Part I **Foundations**	1 - Classical Text Retrieval
	2 - Performance Evaluation
	3 - Advanced Text Processing
Part II **Search Systems**	4 - Index for Text Retrieval
	5 - Semantic Search
	6 - Vector Search
	7 - Retrieval Augmented Generation
Part III **Web & Multimodal**	8 - Web Search
	9 - Content Analysis
	10 - Visual Features
	11 - Audio Features
	12 - Video & Structural Features
Appendix	*Machine Learning (not part of exams)*

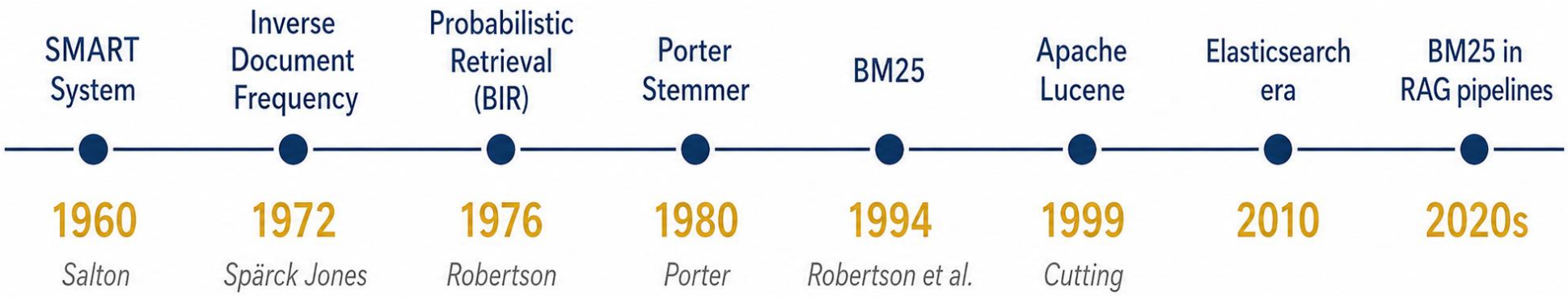
GitHub: https://github.com/roger-weber/mmir-unibasel-hs26
Book: https://roger-weber.github.io/mmir-unibasel-hs26/book/
Quiz: https://roger-weber.github.io/mmir-unibasel-hs26/quiz/

Why is Information Retrieval so important?

- The most-used AI tool you never notice
- Same problem for 60 years:
  - short query → most relevant items in a huge collection

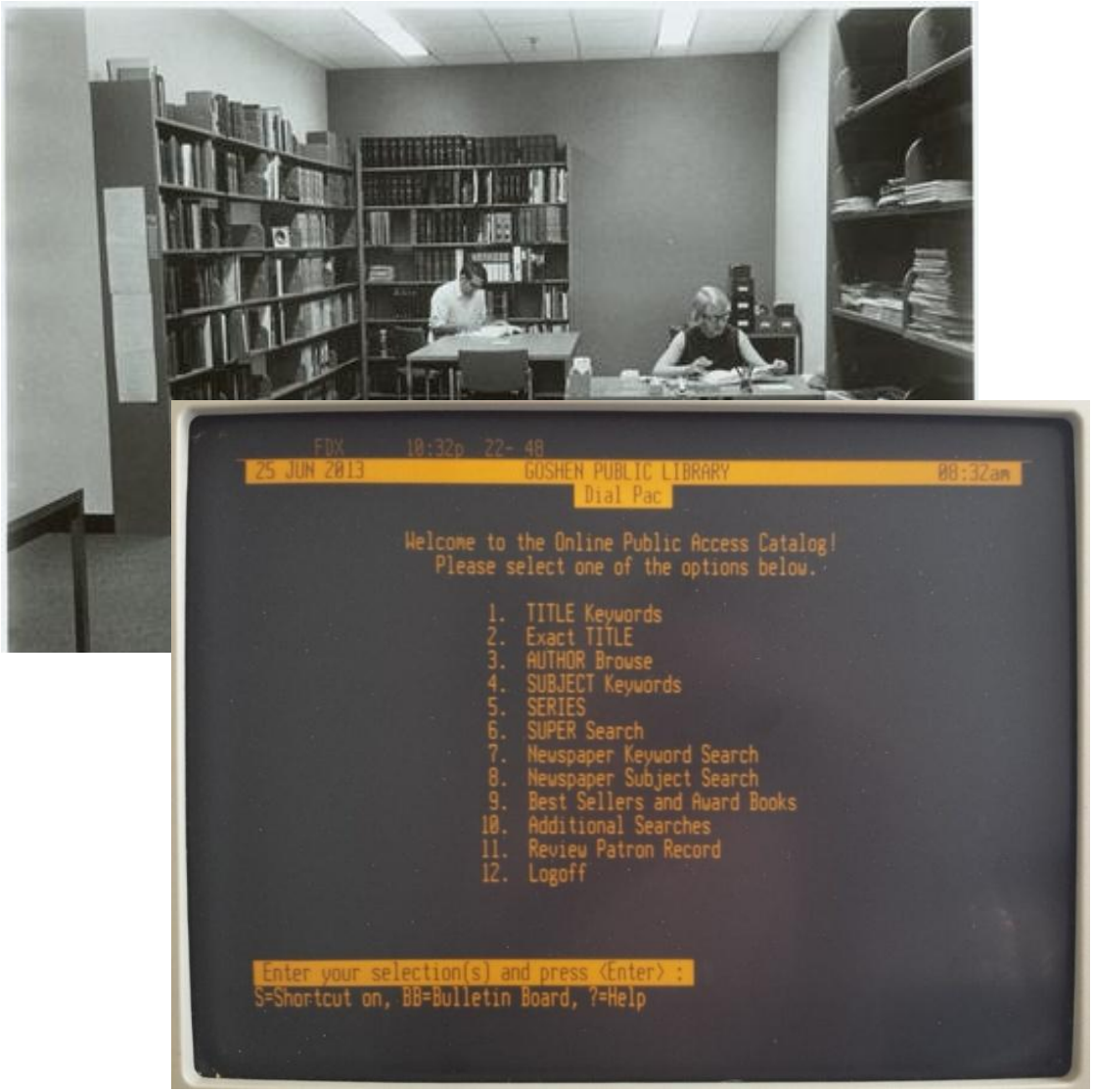
Every era declared search solved; every era then needed it more.

  - over the years, collections exploded and added new challenges (multimedia, quality, distractors)



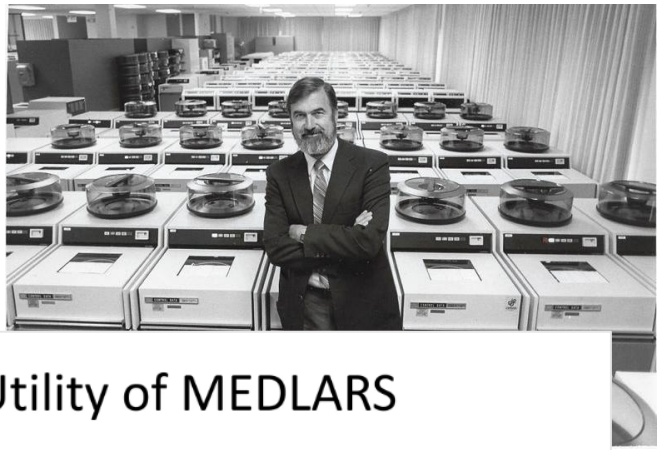
The 1960s: Foundations of Knowledge Management

- 1960s: boom in computerized information retrieval; based on centuries of manual cataloging
- Calvin Mooers coined "information retrieval" (1950); serious R&D began in 1960s
- Most work remained experimental; few machine-readable texts limited adoption
- **Researchers established core metrics: recall and precision**
- IBM developed STAIRS (Storage and Information Retrieval System) for text datasets
- Gerard Salton at Cornell developed SMART — foundational to modern search



The 1970s: The Rise of Computational Models

- 1970s — turning point: retrieval systems became practical
- By end of decade most printed material passed through computer input stages
- **Rise of classical retrieval models**
- Gerard Salton — vector space model: documents/queries as vectors; measure similarity
- Stephen E. Robertson, Karen Spärck Jones — probabilistic retrieval framework



Utility of MEDLARS

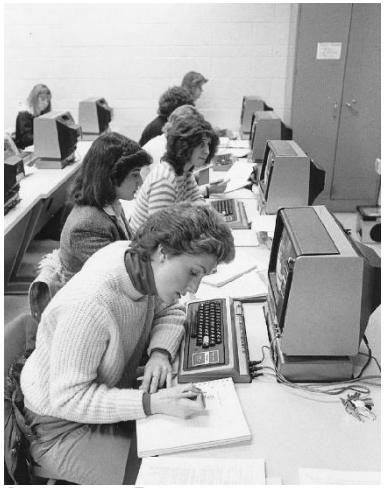
- : MEDLARS is known as the biggest bibliographical database of international level.
- It is important not only for the experts of medical scie
sociologists,
- It has becor
scope is ver
- It also has b
are not conc

INDEXING

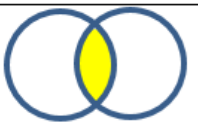
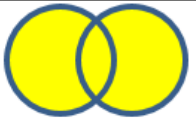
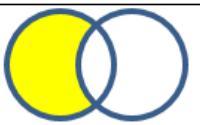
- The MEDLARS data base is created by indexers
- who analyze and describe the contents of journal articles by means of carefully selected terms.
- These terms, called ***subject headings*** or main headings, are contained in an authorized list.

The 1980s: Probabilistic Models and Boolean Systems

- 1980s retrieval refinement and adoption
- Boolean retrieval dominant; AND OR NOT
- Probabilistic retrieval; Binary Independence Retrieval; Probability Ranking Principle
- **Okapi system; BM25 ranking function**
- Latent Semantic Indexing; singular value decomposition



How to Search Using Boolean Operators:

Concept	Search Examples	Results
AND	politics **AND** media children **AND** poverty "civil war" **AND** Virginia	 Results will include both terms
OR	"law enforcement" **OR** police labor **OR** labour 60s **OR** sixties	 Results will include one or both terms
NOT	"civil war" **NOT** American Caribbean **NOT** Cuba therapy **NOT** physical	 Excludes results with the term following NOT

The 1990s: The Web Revolution and Search Engines

- 1990s World Wide Web expansion
- Early search engines: Archie, Gopher, WebCrawler
- Yahoo! directory-based search
- AltaVista full-text and multimedia indexing
- Machine learning, neural networks, relevance feedback

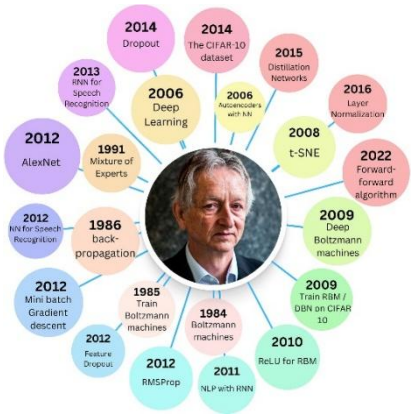




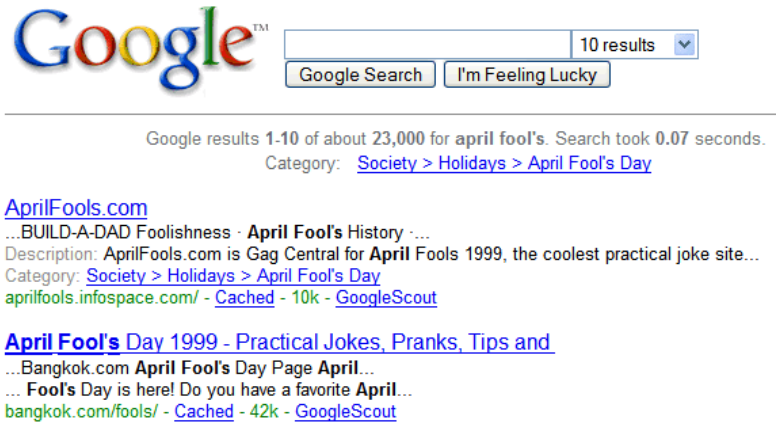


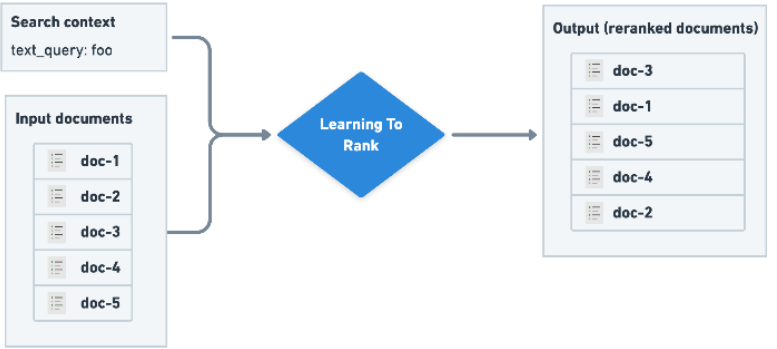
The 2000s: Machine Learning and Ranking Algorithms

- **Google PageRank dominance; user behavior personalization**
- 2000s advanced machine learning in IR
- **Learning to Rank supervised relevance**
- BM25 standard ranking function
- Deep learning resurgence; GPUs, large datasets
- Deep belief networks, convolutional neural networks



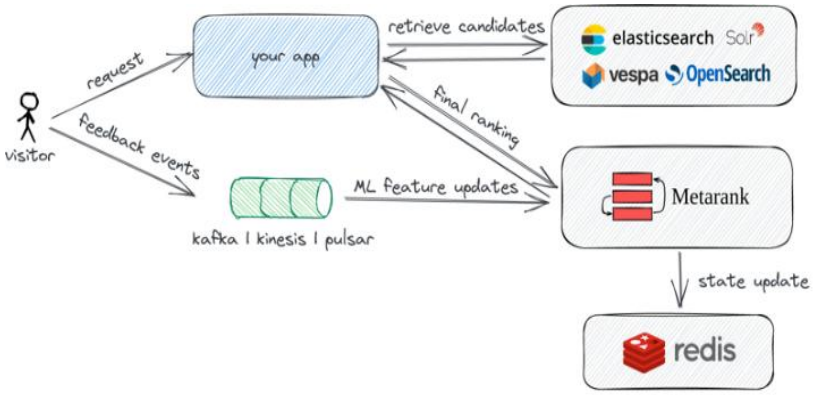
Geoffrey Hinton
Nobel Prize in Physics 2024





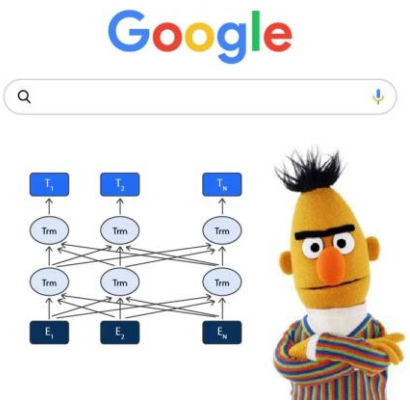
The 2010s: Deep Learning and Neural Information Retrieval

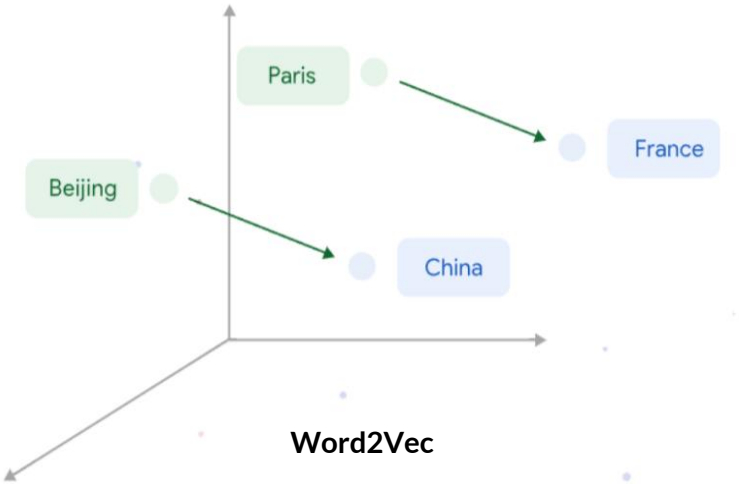
- 2010s deep learning in information retrieval
- Neural ranking models; learned representations
- **Bridge vocabulary gap with learned semantic relationships**
- Google BERT 2018; contextual query understanding



https://opensearch.org/blog/ltr-with-opensearch-and-metarank

– What was Albert Einstein famous for?

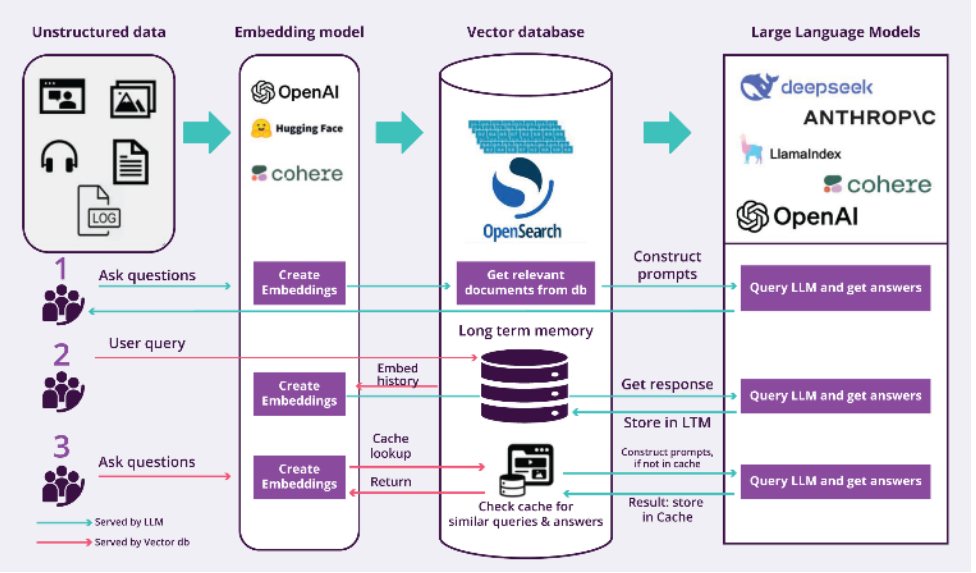




Word2Vec

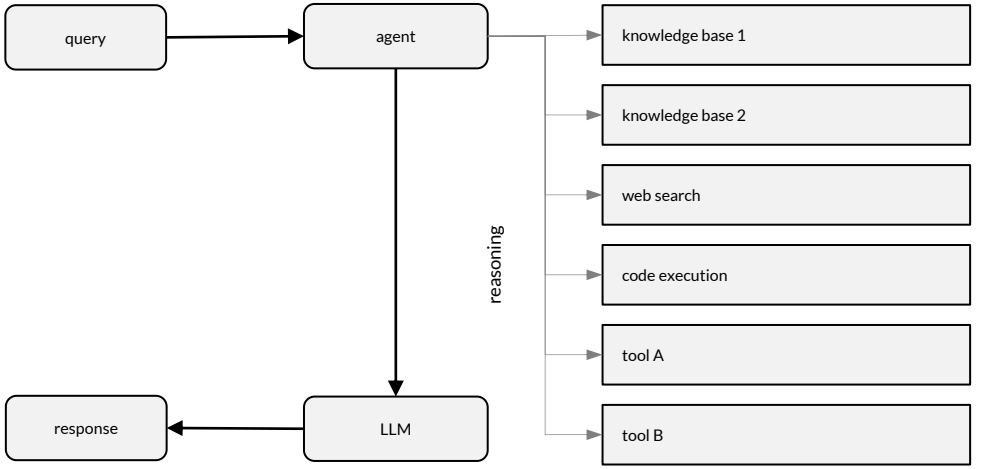
The 2020s: Semantic Search and Vector Databases

- 2020s semantic search; vector-based retrieval
- **Transformer models capture context and meaning**
- Vector databases enable high-dimensional similarity search
- Retrieval-Augmented Generation reduces hallucinations
- Agentic IR multi-step reasoning and intelligent intermediaries



https://opensearch.org/platform/vector-engine

- Galaxus: Abfallkübel
- Amazon: Abfallkübel
- Who played bass in the band that released the song yellow submarine?



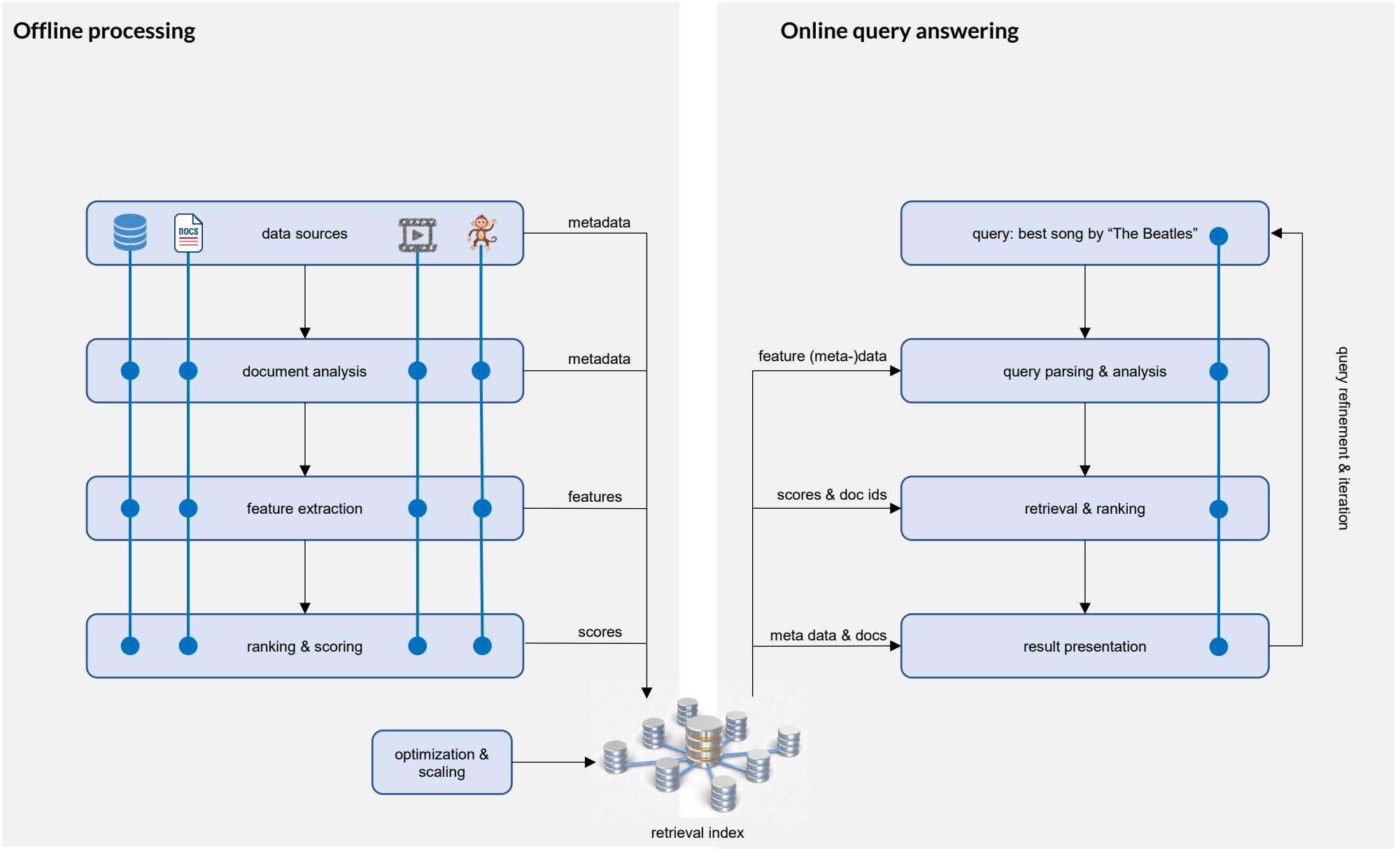
Semantic Gap - why text always was easier

- Multimedia search harder than text
- **Semantic gap**: mismatch between query and media
- Text matches keywords directly
- Images, audio, video need translation
- No fixed relation between keywords and content

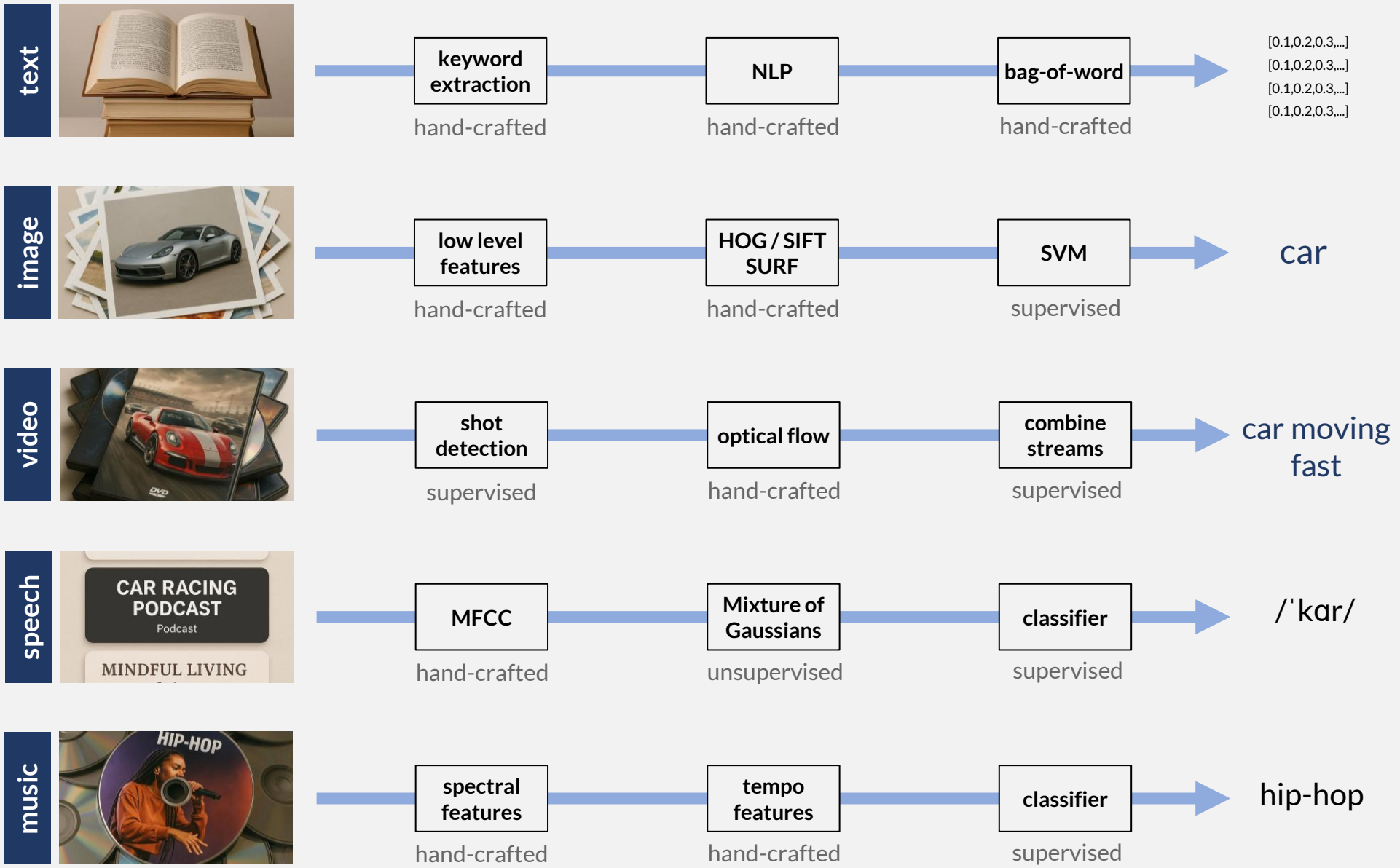
The **semantic gap** refers to the disparity between low-level features extracted from multimedia data and the high-level semantics that humans associate with that data.



How Retrieval Systems work (Generic Architecture)



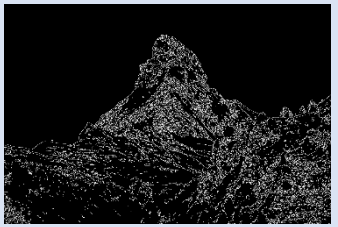
Traditional, mostly hand-crafted pipelines for feature extraction



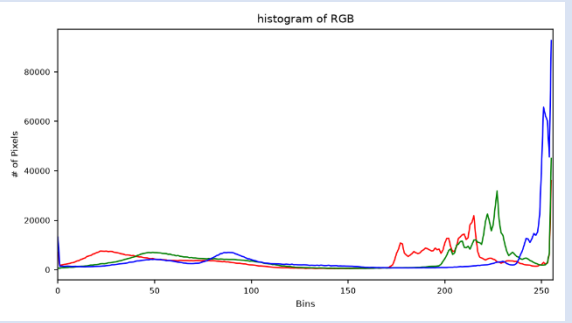
Transformers (multi-modal) help to close the semantic gap

traditional feature engineering









universal prompt engineering

The image centers on a single, dramatic mountain peak with a sharp, pyramid-like summit. Its steep faces are streaked with snow and dark gray rock. The peak looks like the Matterhorn in the Swiss Alps, a shape recognized around the world. It rises well above the surrounding landscape and dominates the frame. The photo was taken from a lower alpine valley or hillside. In the foreground and midground, green slopes are covered with dense evergreen forest, mostly spruce and larch. On the right, a rocky cliff face juts out of the hillside, with trees clinging to its ledges and grassy patches between the stones. On the left, a lower ridge carries scattered patches of leftover snow and a few small structures near its crest. Behind everything, the mountain stands against a pale sky. The palette is cool and natural: The sky is soft powder blue, with wispy white clouds drifting around the summit. The mountain is mostly white snow broken up by slate-gray and charcoal rock ribs and gullies. The lower slopes are deep forest green, with lighter yellow-green meadows in between. The cliff on the right is warm brownish-gray. The light is diffuse and even, probably from a partly cloudy or hazy day. It gives gentle contrast without harsh shadows, and it brings out the texture of the snow and rock. There are no people or animals visible. The living elements are the conifers, which range from tiny dots on the far slopes to a large, detailed branch of …

Generative AI writes the answer, retrieval keeps it anchored to evidence

Knowledge Base: INF-421 uses hybrid retrieval combining **BM25 keyword retrieval** and **dense vector retrieval**. It retrieves the **top 20 documents** from each method and merges them using **Reciprocal Rank Fusion (RRF)**.

Bot A: LLM without knowledge base

- **Student:** What retrieval architecture is used for the INF-421 information system?

  Bot: INF-421 likely uses a **hybrid retrieval architecture**, combining traditional keyword-based retrieval such as **BM25** with **dense vector retrieval**. This is a common approach as it combines exact keyword matching with semantic similarity.

- **Student:** Is it true that it only uses vector search for fast semantic reasoning?

  Bot: Yes, vector search is most likely the main approach used for fast **semantic reasoning in INF-421**. Dense vector retrieval is particularly useful for semantic search because it can identify documents based on meaning rather than exact keyword matches.

- The model contradicted its previous answer, but **smoothly and confidently.** Rather than:
  - "I don't have enough information to verify that."
- **Where did the evidence for "vector search" come from?** There wasn't any.
  - The model generated a plausible answer from its learned knowledge

Bot B: LLM with knowledge base

- **Student:** What retrieval architecture is used for the INF-421 information system?

  Bot: INF-421 uses a **hybrid retrieval architecture** combining **BM25 keyword retrieval** and **dense vector retrieval**. It retrieves the top 20 documents from each method and merges the results using **Reciprocal Rank Fusion (RRF)**. [1]

- **Student:** Is it true that it only uses vector search for fast semantic reasoning?

  Bot: No. The INF-421 retrieval system does **not** use only vector search. According to the retrieval policy, it uses a **hybrid approach combining BM25 keyword retrieval and dense vector retrieval**. The system retrieves the top 20 documents from each method and combines their results using Reciprocal Rank Fusion (RRF). [1]

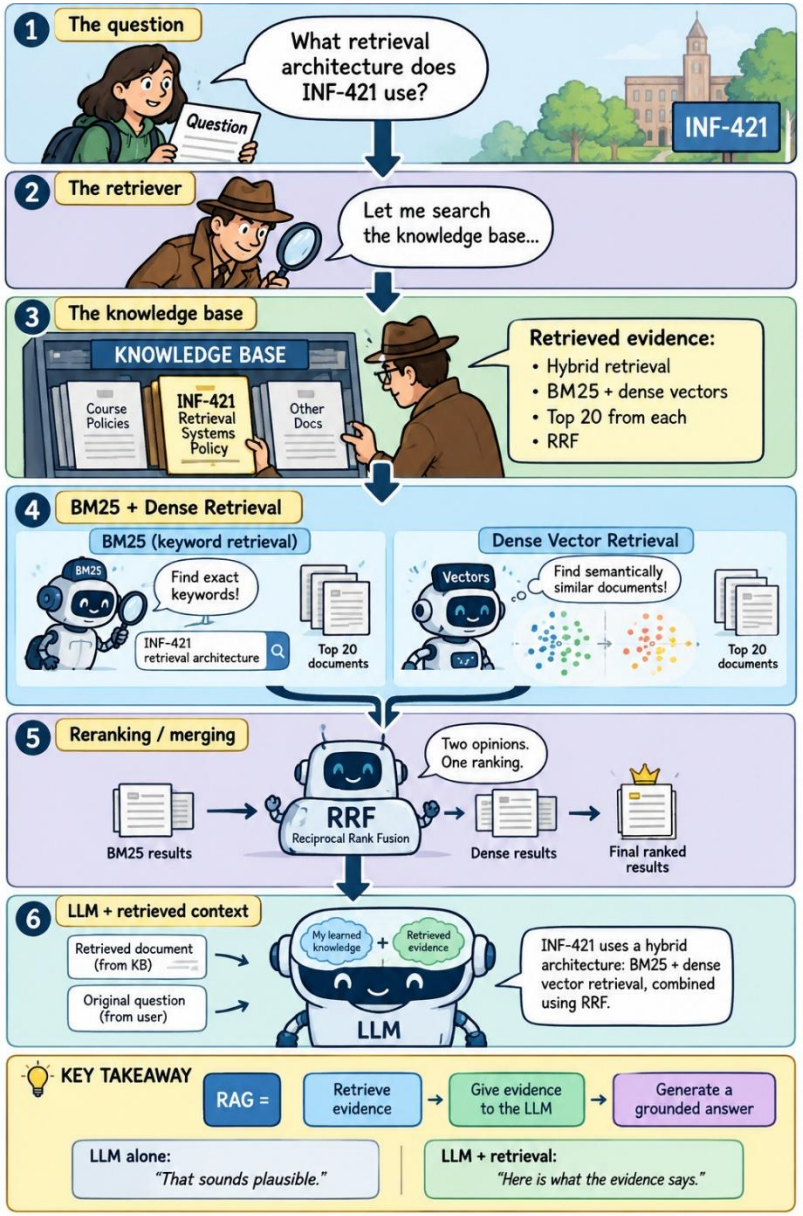
- The model is now **grounded in evidence** and can correct your wrong assumptions.
  - It can still produce so-called **"hallucinations"**.
  - Bots often provide links to the source material ([1]), allowing users to verify the claims.
  - If the relevant information cannot be found, Bot B may fall back to answers similar to those from Bot A.

Sixty years of progress, and nothing was thrown away

- LLMs don't know everything, and go stale
- Retrieval grounds them
- The trust layer: citations, provenance
- Scales where generation can't

- Cumulative, not disposable:
  - precision/recall, BM25, embeddings, all still in use

- Each era adds a layer for a harder problem:
  - semantic gap in text → embeddings
  - semantic gap in images/audio/video → multimodal features
  - scale → indexes, ANN, vector databases
  - from result lists → direct answers & summaries (RAG)



Schedule

Date	Theory: 15:15 / 16:15	Practice: 17:05	Where
Sep 18	1 - Classical Text Retrieval	Ex 1 (new)	University
Sep 25	1 - Classical Text Retrieval / 2 - Performance Evaluation	Ex 2 (new)	University
Oct 2	2 - Performance Evaluation / 3 - Advanced Text Processing	Ex 1 (discuss), Ex 3 (new)	University
Oct 9	3 - Advanced Text Processing	Ex 2 (discuss)	University
Oct 16	4 - Index for Text Retrieval	Ex 3 (discuss), Ex 4 (new)	University
Oct 23	5 - Semantic Search	Ex 4 (discuss), Ex 5 (new)	University
Oct 30	6 - Vector Search	Ex 5 (discuss), Ex 6 (new)	**Zoom**
Nov 6	7 - Retrieval Augmented Generation	Ex 6 (discuss), Ex 7 (new)	**Zoom**
Nov 13	8 - Web Search	Exam Prep, Ex 7 (discuss), Ex 8 (new)	**Zoom**
Nov 20	9 - Content Analysis	Ex 8 (discuss), Ex 9 (new)	University
Nov 27	*No Lessons (Dies Academicus, last Friday in November)*		
Dec 4	10 - Visual Features	Evaluation, Ex 9 (discuss), Ex 10 (new)	University
Dec 11	11 - Audio Features	Ex 10 (discuss), Ex 11 (new)	University
Dec 18	12 - Video & Structural Features	Eval & Prep Exam, Ex 11 (discuss), Ex12 (new)	University